天津理工大学

(2022)

22 1

20220397

2025/09/15-2025/09/20

2025 9 28

-
-
-
-
-
-
-

[32°07’41”N 112°14’59”E]

2020

[32°08’10”N 112°11’10”E]

2003 3000 20 AGV

[32°09’01”N 112°10’28”E]

CTP 100 200 2

[32°08’41”N 112°13’30”E]

600

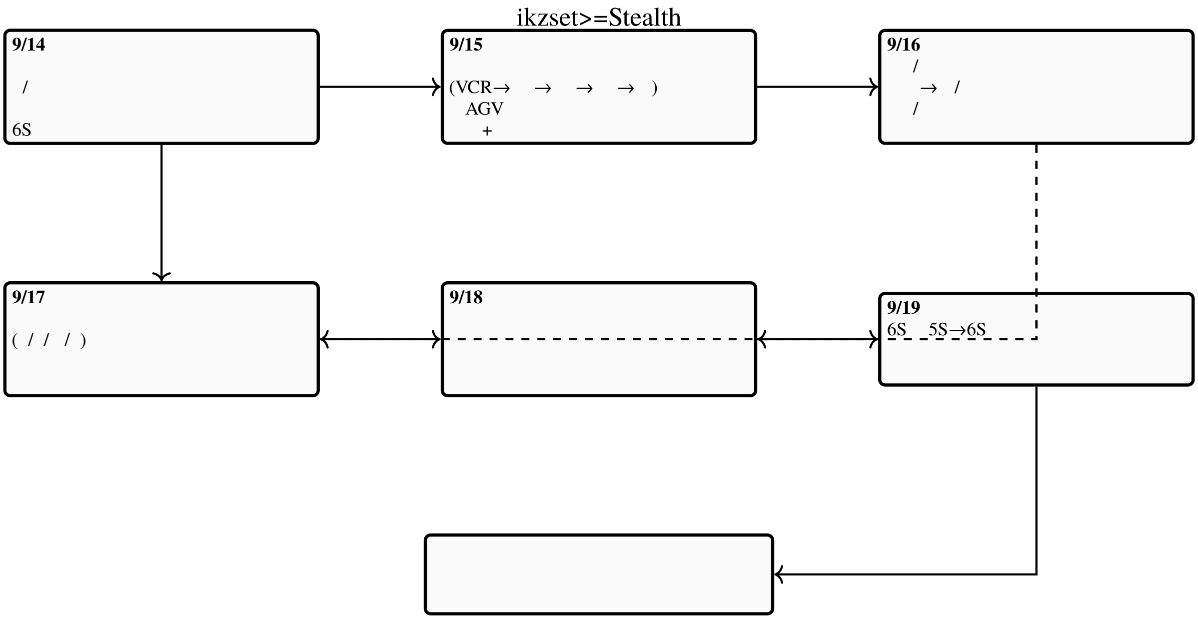
601311.SH

[32°07’50”N 112°15’03”E]

2008

[32°06’21”N 112°12’26”E]

[32°10’08”N 112°09’49”E]



1.

2.

1. **—2025/09/14 14:30**

(1)

1.1.

1.2.

1.3.

1.4.

1.5.

1.6.

1.7.

1.8.

1.9.

1.10.

1.11.

1.12.

(2)

1.1.

1.2.

1.3.

1.4.

1.5.

1.6.

1.7.

1.8.

1.9.

1.10.

1.11.

2. **—2025/09/14 15:00**

(a)

(b)

-
-
-

(c)

(d)

-
-

3. Takafuji

(a)

(b)

(c)

4. **6S** 6S

(a) **Seiri**

(b) **Seiton**

(c) **Seiso**

(d) **Seiketsu**

(e) **Shitsuke**

(f) **Safety**

6S “ ”

1.

2.

—— —— —— —— / / —— EOL, End-Of-Line

- /
- **AGV** “ ” AGV WMS/MES
- / CAN MES
- ABB 5CARA AGV

2 3 “ ”

1. CTP

→ → RIP →

1. /

2. RIP

3. CTP

4.

5.

6. 2–3

2. BB

1.

2. /

3.

4. /

5. “ ”

3.

1. CMYK

2. /

3.

4. /

5.

4. TKS

1.

2.

3.

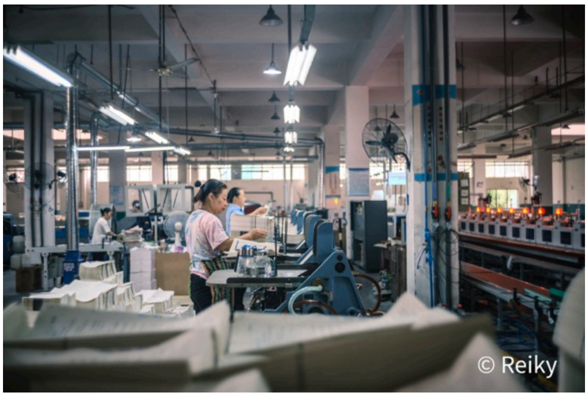
4. “ / ”

5. /

6.

5.

→ → 3



2 3

1.

→ / → / → → → →

2.

1. “ ”

2.

3.

4. “ ”

5.

6. / 120,000

3.

1. /

2.

3.

4.

5.

4.

1. 5,000

2. /

3. 8 / “ 98.5%”

4.

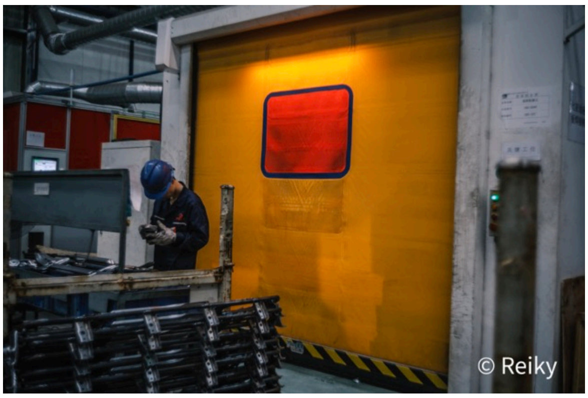
5.

5.

- LFP 250 Wh/kg NCM 180 Wh/kg LFP
- 2000–5000 mAh
- “ 98%”“ >95%” → → → → →

6.

3



1. 9/18

1. / “ — — ”

2. / /

3. TCP 4

4.

5. CAD → → →

2. 6S 9/19

1. “ ”

2. 5S→6S

3. “ ”

4. +

5. 6S “ ” “ ”

/

3.

→ / → → / → + → / / → →

4.

(1) 5000 rpm DIN 6

(2) CZL-16N m=0.3–2.0 1500 /

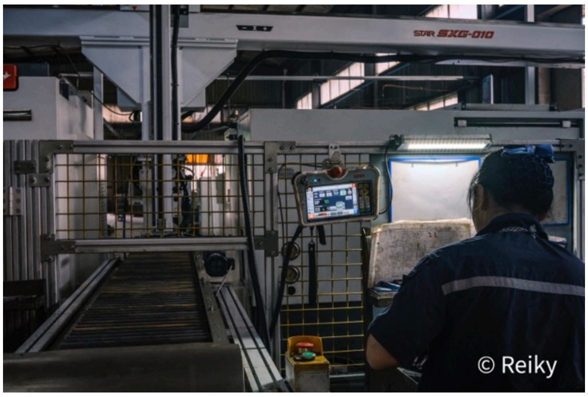
(3)

(4) 58–62 HRC /

(5)

5.

- /
-
-
-
-



/ /

1.

/ → → / → → / → / →

2.

(1) / 120 / / ±0.02 mm

(2) “ + ”

(3) / 18650 21700

(4) 15,000 / 48 8

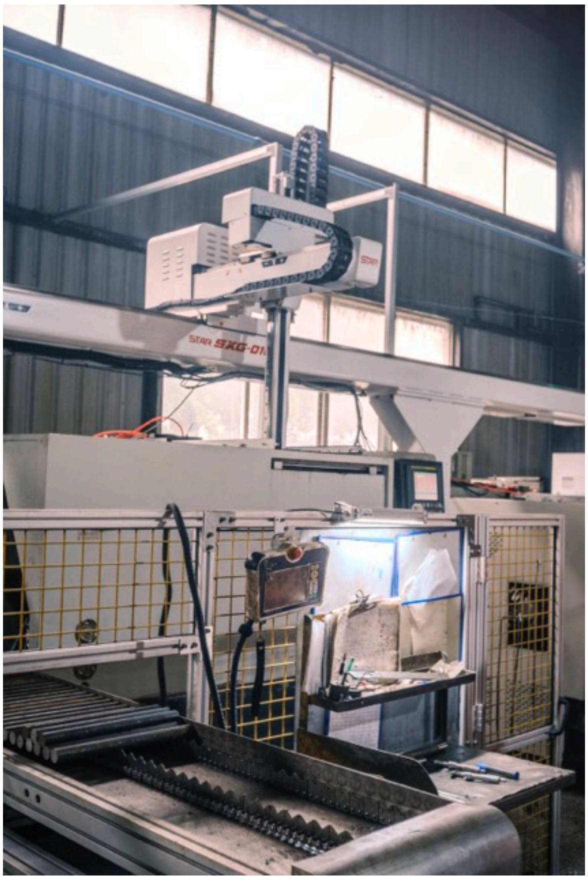
(5) / → → → / → 30 /

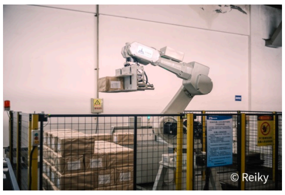
3.

-
-
-
-
-

4.

3

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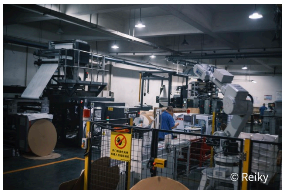
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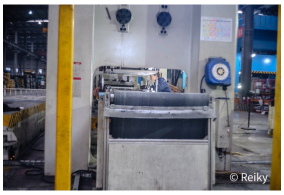
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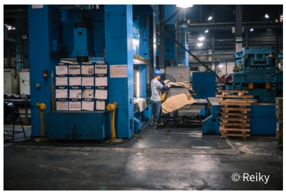
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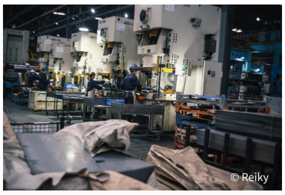
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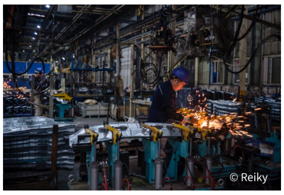
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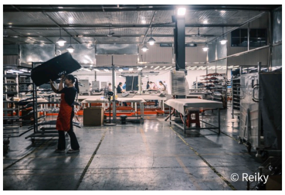
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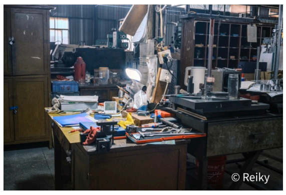
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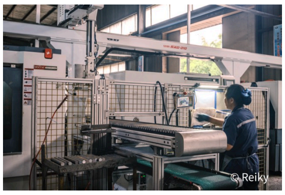
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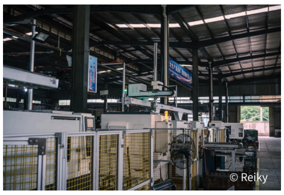
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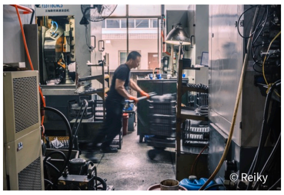
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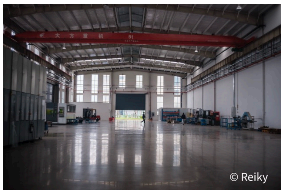
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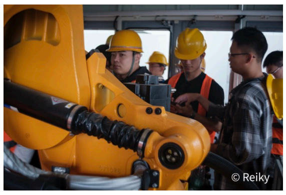
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3 2 /

1. CTP

(1) / RIP CTP / / / /

(2)

- 2–3 min
- 1%–99%
- 10–20 μm

(3)

(4)

(5) ICC

2. BB

(1) / / /

(2) 30,000–45,000 / < 0.2 mm < 60 s

(3)

(4)

(5)

3. /

(1) / / /

(2) 15,000–20,000 / ΔE ΔE < 3 (TVI)

(3) CIP3/CIP4

(4)

(5) /ΔE

4. TKS

(1) + / CMYK×2

(2) 80,000–100,000 / 60k–90k / 4.8–6.0 /

(3) /

(4) OEE / / AI

(1) / / / AGV / / /
→ → / → → /

(2) 60–70 s/ 65 s 16 h/ 300 / $\approx \frac{16\times3600}{65}\times300\approx2.66\times10^5$ “ 20
” /

(3) / + / + / ±0.2 mm

(4) AGV SLAM 300–500 kg ≈8 h / JIT

(5) CAN/LIN / NVH /

(6) + AGV / OEE

(7) / / AGV / /

(8) + AGV + RFID + WIP /

1.

(1) + + / /

(2) ＋

- φ400 mm
- 600 mm
- DIN 5
- 30–40%
- 4,000 rpm

(3) /

(4) /

(5) + + - / MES

1.

(1) / / / /β

(2) ＋

- 120k /
- ±2–3 μm
- < ±1.5%
- NCM/LFP

(3)

(4) / /

(5) DOE + / SPC

2.

(1) /

(2) ±0.1 mm ppm

(3) +

(4)

(5) +

3.

(1) /

(2) 15k ≈8 h ≈98.5% 72 h

(3) SOH

(4) +

(5) CV dV/dt

4.

(1) / → → → / / →

(2) 98% >95%

(3) ESG

(4) /

(5)

5. OEE

$$\text{OEE} = A \times P \times Q, \quad A = \quad , \ P = \quad , \ Q =$$

$A = 0.90, P = 0.92, Q = 0.985 \Rightarrow OEE \approx 0.817$ OEE 18%

(1) A P Q

- /
- / /
- / / SPC +
- ESG /
- / /
- AGV / / MES + /
- OEE DOE +
- /

(1) ΔE

4 2 3 “ / ”[1]

“ / / / ”

1. → → → “ + ”
2. OEE
3. / — “ vs ”
4. → AGV SPC → OEE /
5. “ ” / “ ”

1 TabularX

/				/
	→ →	/	ΔE	/
CTP	RIP→ → →	/		
→	→ → → → →			+ SPC
	→ → → → →EOL	AGV		
	→ → → →			/
	→ → → →			

1: TabularX

-
- VOC
-

1.
2. AGV MES
3.

1

“ — — ” “ / ” “ / ”

1. E-S-D-R Event / State / Decision / Result

- (E):
- (S): SPC
- (D): / / /
- (R): OEE

/

2. “ ” “ + ” / /

3. / + /

4. / /

5. “ ”

6.

7. **ROI** / vs

8. “ — — ” / /

9. “ / / / + playbook” “ → → → ”

“ → → → ” / / KPI

(1–2)		>90% 30%
(1–2)	2 / / OEE	>8 OEE 100%
(1–2)	/ →	50% >80%
(3–6)	+AGV DOE +	<5% 15%
(6+)	(/)	30%

2:

(1) (1) (2) (3) (4) (5)
/OEE/

(2) → / → → →

1. MES PLC
2. OEE
3. “ ” “ ”
4. VOC

(1) → → → “ / ” “ / ”